

Date	07 July 2026	Team Name	anupriya
Project Name	CaptionAI	Maximum Marks	1 Mark

SCALABILITY & FUTURE PLAN

SmartBridge Skill Wallet Submission

This document outlines the strategic roadmap for scaling the CaptionAI platform to accommodate growing user traffic and lists key functional enhancements planned for upcoming production releases.

Scalability Analysis

SCALABILITY ASPECT	DETAILS
Current Architecture	Single-instance Flask local application server serving a unified front-end interface, communicating directly with Groq API endpoints via secure .env keys over localized Ngrok public tunnels.
Scalability Approach	Migrate the Flask core engine into Docker container instances deployed onto multi-zone cloud nodes behind an Nginx load balancer, implementing an asynchronous task queue (like Celery with Redis) to prevent Groq API timeout blocks under concurrent traffic loads.
Expected Growth / Load	Staged implementation to transition seamlessly from low-volume mock testing requests up to thousands of active generation calls monthly, scaling horizontally to keep overall processing speeds under 3 seconds per request block.

Future Roadmap

S.NO	PLANNED FEATURE / ENHANCEMENT	TARGET TIMELINE	PRIORITY
1	Multi-Language Generation Engine Add localized native language prompts for global marketing regions.	Q3 2026	High
2	Vision-Based Image Input Tool Integrate Llama vision models to extract prompt data straight from uploaded graphics.	Q4 2026	Medium
3	User Profiling & Persistent Dashboard History Implement persistent relational database tracking to securely store historical content logs.	Q1 2027	Medium

S.NO	PLANNED FEATURE / ENHANCEMENT	TARGET TIMELINE	PRIORITY
4	Automated Cross-Platform Social Posting API Direct integration with social platform publishing networks to post content directly from the UI.	Q2 2027	Low